

Jingchuan Chen

Ph.D.

Department of Atmospheric Sciences

Texas A&M University, College Station, Texas, United States

E-mail: jingchuanchen@tamu.edu

TEL: +1 (979) 326 0954

PROFESSIONAL POSITIONS

-
- **Post Doctoral Research Associate in Atmospheric Sciences, Texas A&M University** College Station, TX, USA
Advisor: Dr. Sarah D. Brooks Jan. 2025 - Present
Research: *The Development and Comparison of Contact Freezing Measurement Techniques*

EDUCATION

-
- **Ph.D. in Environmental Sciences, Peking University** Beijing, China
Advisor: Dr. Zhijun Wu Aug. 2018 - Jul. 2024
Thesis: *Ice Nucleation Activity of East Asian Dust Aerosols and Its Influencing Factors*
 - **B.S. in Environmental Sciences, Jilin University** Changchun, China
Advisor: Dr. Yang Ning; *Graduated with Honors* Sep. 2014 - Jul. 2018
Thesis: *Distribution Characteristics of Atmospheric Ice-Nucleating Particles in Beijing*

SELECTED AWARDS & SCHOLARSHIPS

-
- Award for Scientific Research, Peking University 2022
 - **Grand Prize**, 30th “Challenge Cup” Peking University May Fourth Youth Science Award Competition 2022
 - Merit Student (**Top 12%**), Peking University 2021
 - Outstanding Graduate (**Top 3%**), Jilin University 2018
 - First-Class Scholarship (**Top 7%**), Jilin University 2018
 - Fude Scholarship (**Top 5%**), Jilin University 2017
 - National Scholarship (China) (**Top 1%**) 2017
 - Outstanding Student, Jilin University 2016

PROFESSIONAL ACTIVITIES & SERVICE

-
- **Early Career Convener.** American Geophysical Union Annual Meeting (AGU26)
Session: *Atmospheric Aerosols and Their Interactions with Clouds, Radiation, and Climate* 2026
 - **Peer Reviewer.** *Environmental Science & Technology, Journal of Geophysical Research: Atmospheres, Atmospheric Chemistry and Physics, Atmospheric Measurement Techniques, Atmospheric Environment, Atmospheric Research, ACS Omega* Ongoing
 - **Lab Manager.** College of Environmental Sciences and Engineering, Peking University
Co-managed laboratory operations, including safety oversight, instrument training, and routine instrument maintenance. 2021-2024
 - **Mentor.** Center for Atmospheric Chemistry and the Environment, Texas A&M University
Mentored graduate student in C-ICE experimental methods and data processing. Ongoing
College of Environmental Sciences and Engineering, Peking University
Trained and mentored 6 graduate students in ice nucleation array operation and data analysis. 2019-2024
 - **Conference Coordinator.**
The 6th International Conference on Environmental Simulation and Pollution Control, Beijing, China 2019

PEER-REVIEWED PUBLICATIONS

(Citation Index h = 9; Citations = 234; [Google Scholar](#) & [ORCID: 0000-0002-5200-982X](#))

First-author Papers

- [1] **Chen, J.**; Xu J.; Wu, Z.*; ... Hu, M., Decreased dust particles amplify the cloud cooling effect by regulating cloud ice formation over the Tibetan Plateau. *Science Advances* 2024, 10(37): eado0885.
- [2] **Chen, J.**; Wu, Z.*; Meng, X.; ... Hu, M., Observational evidence for the non-suppression effect of atmospheric chemical modification on the ice nucleation activity of East Asian dust. *Sci. Total Environ.* 2023, 861, 160708.
- [3] **Chen, J.**; Wu, Z.*; Chen, J.; ... Hu, M., Size-resolved atmospheric ice-nucleating particles during East Asian dust events. *Atmospheric Chemistry and Physics* 2021, 21, (5), 3491-3506.

Selected Co-author Papers

- [1] Meng, X.; Wu, Z.*; **Chen, J.**; ... Hu, M., Particle phase state and aerosol liquid water greatly impact secondary aerosol formation: insights into phase transition and its role in haze events. *Atmospheric Chemistry and Physics* 2024, 24, (4), 2399-2414.
- [2] Qiu, Y.; Qiu, T.; Wu, Z.*; ... **Chen, J.**; ... Hu, M., Observational Evidence of Brown Carbon Photobleaching in Urban Atmosphere at Molecular Level. *Environ. Sci. Technol. Lett.* 2024, 11, (10), 1032-1039.
- [3] Chen, L.; Peng, C.; **Chen, J.**; ... Tang, M.*, Effects of heterogeneous reaction with NO₂ on ice nucleation activities of feldspar and Arizona Test Dust. *J. Environ. Sci.* 2023, 127, 210-221.
- [4] Qiu, Y.; Wu, Z. *; ... **Chen, J.**; ... Hu, M., Secondary aerosol formation drives atmospheric particulate matter pollution over megacities (Beijing and Seoul) in East Asia. *Atmospheric Environment* 2023, 301, 119702.
- [5] Zhang, C.; Wu, Z.*; **Chen, J.**; ... Kanji, Z., Ice-nucleating particles from multiple aerosol sources in the urban environment of Beijing under mixed-phase cloud conditions. *Atmospheric Chemistry and Physics* 2022, 22, (11), 7539-7556.
- [6] Zhao, G.; Zhu, Y.; Wu, Z.; Zong, T.; **Chen, J.**; ... Hu, M., Impact of aerosol–radiation interaction on new particle formation. *Atmospheric Chemistry and Physics* 2021, 21, (13), 9995-10004.
- [7] Chen, J.; Wu, Z.; Zhao, X.; Wang, Y.; **Chen, J.**; ... Li, S., Atmospheric Humic-Like Substances (HULIS) Act as Ice Active Entities. *Geophys. Res. Lett.* 2021, 48, (14).

Selected Patents

- [1] **Chen, J.**; Wu, Z.*, Atmospheric Ice-Nucleating Particle Image Recognition and Data Processing System. [Chinese National Invention Patent](#), CN202211454107.4. Under review.
- [2] **Chen, J.**; Wu, Z.*, Atmospheric Ice-Nucleating Particle Image Recognition and Data Processing Software. [Chinese National Computer Software Copyright](#), 2022SR1386636. Oct. 31, 2022.
- [3] Wu, Z.*; **Chen, J.**; Meng, X., Automated Multi-Stage Single Particle Impact Sampler and Sampling Method. [Chinese National Invention Patent](#), CN202110768513.7. Apr. 26, 2022.

Selected Conference Presentations

- [1] **Chen, J.** et al., A Novel Cold-Stage System for Quantifying Impact-Initiated Contact Freezing of Ice Nucleating Particles. **American Geophysical Union Annual Meeting (AGU25)**, New Orleans, LA, 2025 (Oral)
- [2] **Chen, J.** et al., Dust Particles Regulate Ice Formation of Mixed-Phase Clouds Over the Tibetan Plateau. **American Geophysical Union Annual Meeting (AGU24)**, Washington, D.C., 2024 (Oral)
- [3] **Chen, J.** et al., Vertical Measurements of Atmospheric Pollutants Based on an Aircraft Platform. **2024 Symposium on Synergistic Governance of Pollution Reduction and Carbon Mitigation**, Beijing, China, 2024 (Oral)
- [4] **Chen, J.** et al., Size-Resolved East Asian Dust Ice-Nucleating Particles: Impact of Atmospheric Chemical Modification on Ice Nucleation Activity. **The 104th American Meteorological Society Annual Meeting**, Baltimore, Maryland, 2024 (Oral)
- [5] **Chen, J.** et al., Seasonal East Asian Dust Impacts Ice Formation in Clouds Over the Tibetan Plateau, China: Experimental Evidence of Ice-Nucleating Particles. **The 103rd American Meteorological Society Annual Meeting**, Denver, Colorado, 2023 (Oral)

TEACHING EXPERIENCE

- **Technical Instructor.** Undergraduate Environmental Monitoring Laboratory, Peking University
Guided undergraduate students in aerosol filter sampling and offline chemical analysis, including gravimetric mass determination, ion chromatography (IC), total organic carbon (TOC), and EC/OC analysis. 2020-2021
- **Teaching Assistant.** Peking University
Served as a teaching assistant for three undergraduate courses and one graduate course in atmospheric particle measurement and experimentation. Mar. 2019 - Jul. 2021.

RESEARCH EXPERIENCE

- **Contact Freezing of Ice-Nucleating Particles (INPs)** Jan. 2025 - Present
 - Investigate contact freezing as a potentially important yet poorly constrained pathway of cloud ice formation; in particular, [impact-initiated contact freezing](#) remains understudied despite its direct relevance to particle-droplet collision processes in the atmosphere.
 - Lead experimental research on impact-initiated contact freezing within a multi-method project on INPs, aimed at resolving methodological differences and improving quantitative constraints on atmospheric contact freezing.
 - Conceive, develop, and validate [C-ICE \(Cold-stage system for Impact-initiated Contact freezing Experiments\)](#), a new method for controlled and repeatable investigation of impact-initiated contact freezing.
 - Use C-ICE to identify the key factors governing impact-initiated contact freezing and to establish a quantitative approach for laboratory study of this process.
 - Build benchmark datasets and develop parameterization frameworks for impact-initiated contact freezing to support laboratory interpretation and future model applications.
([First manuscript in preparation; second manuscript in progress.](#))
- **Observations of Atmospheric Ice-Nucleating Particles** May 2018 - Dec. 2024
 - Size-Resolved Atmospheric Ice-Nucleating Particles During East Asian Dust Events** May 2018 - Jul. 2020
 - Clarified the [size dependence](#) of both INP concentrations and surface ice-active-site densities of dust particles.
 - Developed [a new set of parameterizations](#) for ice nucleation activities (INA) based on size-resolved nucleation properties of Asian mineral dust particles (suitable for -35 to -6°C). ([1 paper published](#))
 - Effect of Atmospheric Chemical Aging on the Ice Nucleation Activity of East Asian Dust** May 2018 - Dec. 2024
 - Demonstrated the [non-suppression effect of atmospheric chemical aging on the INA](#) of East Asian dust INPs through integrated analysis of physical and chemical properties of particulate matter collected during two field observations.
 - Designed and led a two-year field observation to investigate the evolution of dust INA during long-range transport over ~1000 km. ([1 paper published; 1 patent](#))
 - Dust Ice-Nucleating Particles and Their Climate Effects over the Tibetan Plateau** Oct. 2020 - Sep. 2024
 - Quantified INP concentrations and their spatial distribution over the Tibetan Plateau (TP), identifying [mineral dust as the dominant INP source](#) in this region.
 - Demonstrated that the springtime dust influxes lead to seasonally elevated ice concentrations in mixed-phase clouds by integrating data from [multi-source observations](#) and [snowpack samples](#) collected from five glaciers.
 - Revealed that the decadal reduction in dustiness from 2007 to 2019 results in decreased springtime dust INPs, thereby [amplifying the cooling effect of clouds](#) over the TP. ([1 paper published; 1 patent](#))
- **Field Campaigns and Vertical Measurements of Atmospheric Pollutants** May 2018 - Dec. 2024
 - Designed and deployed a range of [aerosol sampling devices](#), and conducted vertical and [airborne in situ measurements](#) of atmospheric gaseous pollutants during 7 research flights. (Oct. 2018 - Jun. 2022; [4 patents](#))
 - Coordinated a two-month field campaign in Lulang (3,000 m a.m.s.l.), leading a research team in measurements of aerosols, fog, and precipitation to investigate atmospheric processes over the Tibetan Plateau. (Mar. - May 2021)
 - Participated in 7 field campaigns and developed extensive hands-on expertise in atmospheric aerosol sampling, measurement, and analytical techniques, contributing to 10+ co-authored publications. (May 2018 - Dec. 2024)